

60016 OPERATIONS RESEARCH

Fundamental Theorem of Linear Programming

Lecture 03

Last Lecture

- ▶ Standard form of an LP
 - ▶ Minimisation problem
 - ▶ No inequalities
 - ▶ All equalities have non-negative right hand sides
 - ▶ All variables non-negative

This Lecture

- ▶ Basic solutions
- ▶ Algebra vs. geometry
- ▶ Fundamental theorem of linear programming
- ▶ Basic representations

Assumptions

- ▶ From now on we focus on LPs in standard form

$$\begin{array}{ll}\min & z = c^T x \\ \text{s.t.} & Ax = b \\ & x \geq 0\end{array} \quad (\mathcal{LP})$$

with data $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ (where $b \geq 0$).

- ▶ We assume:
 - ▶ # of variables = $n \geq m$ = # of equations (otherwise, the system $Ax = b$ is overdetermined);
 - ▶ rows of A are linearly independent (otherwise, the constraints are redundant or inconsistent).
- $\Rightarrow$ $\text{rank}(A) = m$

Linear Dependence

Linear dependence of rows in A implies either:

- ▶ **contradictory constraints**

i.e., no solution to $Ax = b$, e.g.

$$x_1 + x_2 = 1$$

$$x_1 + x_2 = 2$$

- ▶ **redundant constraints**, e.g.

$$x_1 + x_2 = 1$$

$$2x_1 + 2x_2 = 2$$

Index Sets

- ▶ Consider only the system of **linear equations** in problem $\mathcal{LP}$.

$$\boxed{A x = b}$$

- ▶ Let $A = [a_1, \dots, a_n]$, where $a_i \in \mathbb{R}^m$ is the **i th column vector** of A .
- ▶ Select a subset of m columns a_i that are **linearly independent**. This is always possible since $m = \text{rank}(A)$ and $n \geq m$.
- ▶ Collect in the **index set** I the indexes for these m columns. I is therefore a subset of $\{1, \dots, n\}$.

Definition: The matrix $B = B(I) \in \mathbb{R}^{m \times m}$ consisting of the columns $\{a_i\}_{i \in I}$ is called the **basis** corresponding to the index set I .

Example: Partition of A

$$A = \begin{bmatrix} 2 & 4 & 3 & 3 & 1 & 0 \\ 3 & -3 & 4 & 2 & 0 & 1 \\ -1 & 2 & 1 & 2 & 0 & 0 \end{bmatrix}$$

Choose $I = \{1, 5, 2\}$

$$\Rightarrow B(I) = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 0 & -3 \\ -1 & 0 & 2 \end{bmatrix}$$

Basic Solutions

Definition: A solution x to $Ax = b$ with $x_i = 0$ for all $i \notin I$ is a **basic solution (BS)** to $Ax = b$ with respect to the index set I .

Definition: A solution x satisfying both $Ax = b$ and $x \geq 0$ is a **feasible solution (FS)**.

Definition: A feasible solution which is also basic is a **basic feasible solution (BFS)**.

Basic Solutions (cont)

Assume for example that $I = \{1, \dots, m\}$.

$$\begin{array}{ccccccc} a_{11}x_1 + \dots + a_{1m}x_m + a_{1,m+1}x_{m+1} + \dots + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + \dots + a_{2m}x_m + a_{2,m+1}x_{m+1} + \dots + a_{2n}x_n & = & b_2 \\ \vdots & & \vdots \\ a_{m1}x_1 + \dots + a_{mm}x_m + a_{m,m+1}x_{m+1} + \dots + a_{mn}x_n & = & b_m \end{array}$$

The following system is equivalent to $Bx_B = b$.

$$\begin{array}{ccccccc} a_{11}x_1 + \dots + a_{1m}x_m & + & a_{1,m+1}0 + \dots + a_{1n}0 & = & b_1 \\ a_{21}x_1 + \dots + a_{2m}x_m & + & a_{2,m+1}0 + \dots + a_{2n}0 & = & b_2 \\ \vdots & & \vdots & & \vdots \\ a_{m1}x_1 + \dots + a_{mm}x_m & + & a_{m,m+1}0 + \dots + a_{mn}0 & = & b_m \end{array}$$

Basic Solutions (cont)

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Basic Solutions (cont)

Observation: The **basic solution** corresponding to I is **unique**.

As the vectors $\{a_i\}_{i \in I}$ are **linearly independent**, the basis B is **invertible**. Thus, the system

$$\boxed{B x_B = b}$$

has a **unique solution** $x_B = B^{-1}b \in \mathbb{R}^m$.

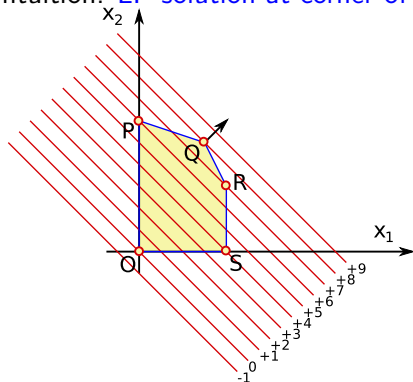
Define $x = (x_1, \dots, x_n)$ through

$$(x_i)_{i \in I} = x_B \quad \text{and} \quad (x_i)_{i \notin I} = 0.$$

This x is **the unique basic solution** to $Ax = b$ **with respect to** I .

Algebra vs. Geometry

- ▶ Geometric intuition: LP solution at corner of feasible set



- ▶ Algebra: Corners of feasible set correspond to basic feasible solutions

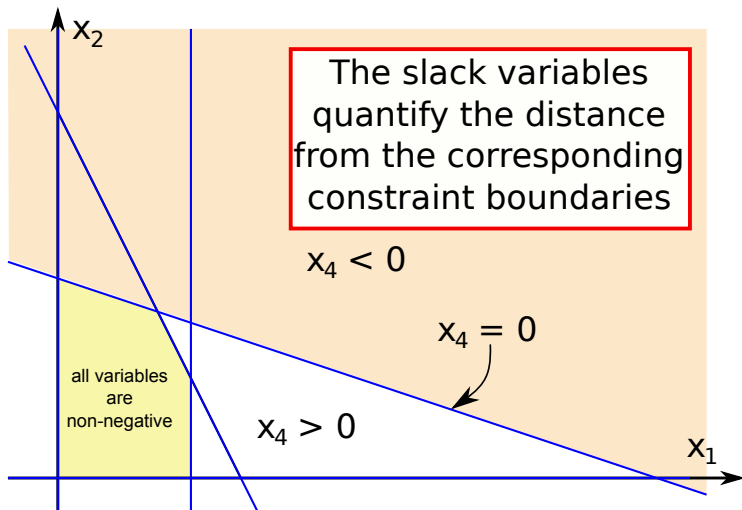
Example 1 (revisited)

$$\begin{array}{ll} \max & y = x_1 + x_2 \quad : \text{objective function} \\ \text{s.t.} & 2x_1 + x_2 \leq 11 \quad : \text{constraint on availability of X} \\ & x_1 + 3x_2 \leq 18 \quad : \text{constraint on availability of Y} \\ & x_1 \leq 4 \quad : \text{constraint on demand of A} \\ & x_1, x_2 \geq 0 \quad : \text{non-negativity constraints} \end{array}$$

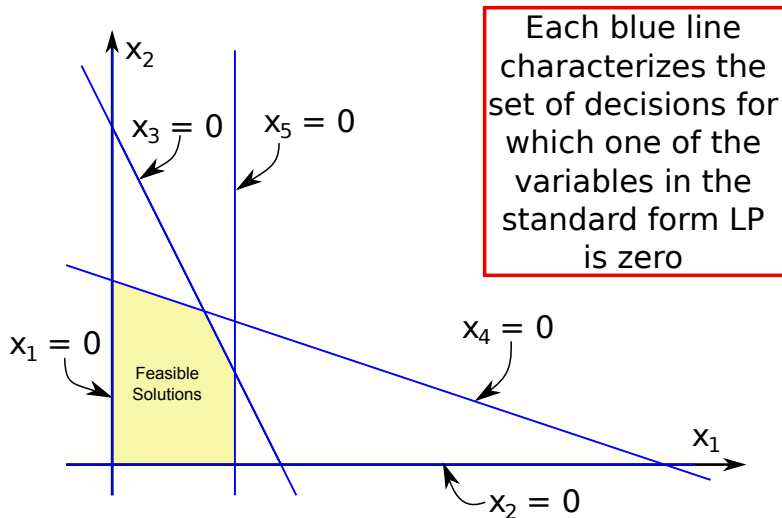
In standard form: $n = 5$ variables & $m = 3$ constraints

$$\begin{array}{ll} -\min & z = -x_1 - x_2 \\ \text{s.t.} & 2x_1 + x_2 + x_3 = 11 \\ & x_1 + 3x_2 + x_4 = 18 \\ & x_1 + x_5 = 4 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{array}$$

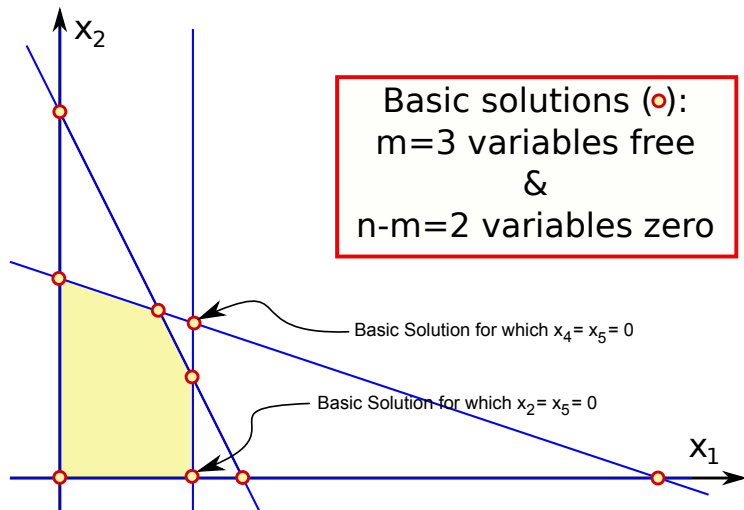
Algebra vs. Geometry (cont)



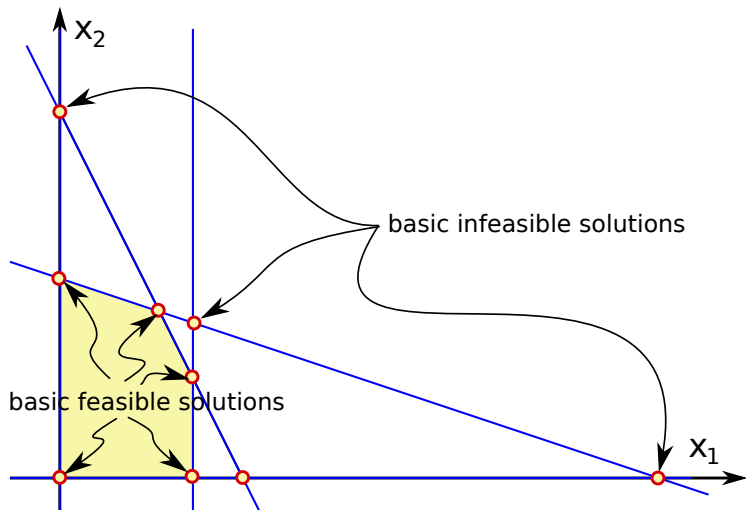
Algebra vs. Geometry (cont)



Algebra vs. Geometry (cont)



Algebra vs. Geometry (cont)



Importance of BFS

Vertices of the feasible set = basic feasible solutions!

- ▶ **Geometry**: optimum always achieved at a **vertex**
- ▶ **Algebra**: optimum always achieved at a **BFS**

Definition: Given an **LP in standard form**, a feasible solution to the constraints $\{A x = b; x \geq 0\}$ that achieves the optimal value of the objective function is called an **optimal feasible solution**. If the solution is basic then it is an **optimal BFS**.

Fundamental Theorem of LP

Theorem 1: For an LP in standard form with $\text{rank}(A) = m \leq n$:

1. $\exists$ a feasible solution $\Rightarrow \exists$ a BFS.
2. $\exists$ an optimal solution $\Rightarrow \exists$ an optimal BFS.

The reverse is in general not true:

- ▶ there may be feasible solutions that are not BFS
- ▶ there may be optimal solutions that are not BFS

The naïve statement “an LP has an optimal BFS” is also in general not true as the LP may be infeasible or unbounded.

Searching for Optima

- ▶ Theorem 1 reduces solving an LP to **searching over BFS's**.
- ▶ For an **LP in standard form** with n variables and m constraints, there are

$$\binom{n}{m} = \frac{n!}{m! (n-m)!}$$

possibilities of selecting m columns in the A matrix.

- ⇒ There are at most $\binom{n}{m}$ basic solutions: **a finite number of possibilities!**
- ⇒ Theorem 1 offers an obvious but **terribly inefficient** way of computing the optimum through a **finite search**.

Number of BFS

Note: There are $\binom{n}{m}$ index sets $I \subseteq \{1, \dots, n\}$ with $|I| = m$.

$\Rightarrow$ The number of distinct BFS is finite and usually $< \binom{n}{m}$
for the following reasons:

1. $B(I)$ may be singular,
2. the BS corresponding to I may not be feasible.

A "Small" Problem

Let $m = 30$, and $n = 100$.

$$\binom{100}{30} = \frac{100!}{30! 70!} \approx 2.9 \times 10^{25}.$$

It takes approximately 10^{12} years if we check 10^6 sets/sec.

(The age of the universe is $\approx 14 \times 10^9$ years!)

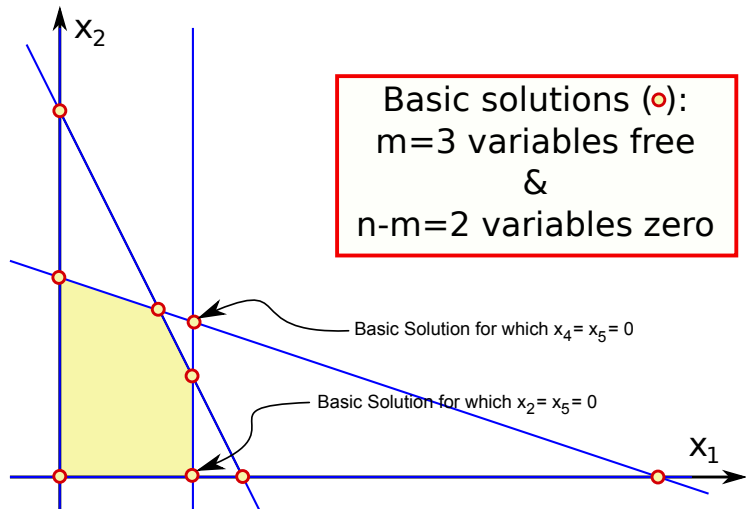
Basic Variables

Fix an **index set** I with $|I| = m$ and $B(I)$ invertible.

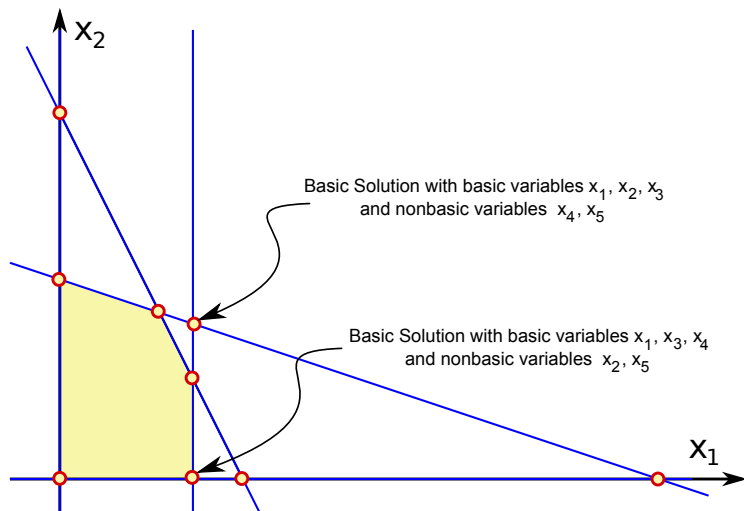
Definition The variables $\{x_i\}_{i \in I}$ are referred to as the **basic variables (BV)**, while the variables $\{x_i\}_{i \notin I}$ are called the **nonbasic variables (NBV)** corresponding to I .

Note: By construction, the **nonbasic variables are always zero**, but the **basic variables can be zero or non-zero**.

Example: Algebra vs Geometry



Example: Basic vs Nonbasic Variables



Basic Representation

Fix an **index set** I with $|I| = m$ and $B(I)$ invertible.

Definition: The **basic representation** corresponding to I is the (unique) reformulation of the system $(z = c^T x, Ax = b)$ which expresses **the objective function value z and each BV as a linear function of the NBV's**:

$$\begin{bmatrix} z \\ x_B \end{bmatrix} = f(x_N),$$

where

- ▶ $x_B = [x_i | i \in I]$ (**BV's**),
- ▶ $x_N = [x_i | i \notin I]$ (**NBV's**) and
- ▶ $f : \mathbb{R}^{n-m} \rightarrow \mathbb{R}^{m+1}$ is **linear**.

Matrix Partition

Let $A = [a_1, \dots, a_n]$, where $a_i \in \mathbb{R}^m$ is the i th column of A . For any index set $I \subseteq \{1, \dots, n\}$ with $|I| = m$. Define

- ▶ $B = B(I) = [a_i | i \in I]$;
- ▶ $N = N(I) = [a_i | i \notin I]$;
- ▶ $c_B = c_B(I) = [c_i | i \in I]$;
- ▶ $c_N = c_N(I) = [c_i | i \notin I]$;
- ▶ $x_B = x_B(I) = [x_i | i \in I]$;
- ▶ $x_N = x_N(I) = [x_i | i \notin I]$.

This implies

$$Ax = Bx_B + Nx_N \quad \text{and} \quad c^T x = c_B^T x_B + c_N^T x_N.$$

Example: Partition of A

$$A = \begin{bmatrix} 2 & 4 & 3 & 3 & 1 & 0 \\ 3 & -3 & 4 & 2 & 0 & 1 \\ -1 & 2 & 1 & 2 & 0 & 0 \end{bmatrix}$$

Choose $I = \{3, 4, 5\}$

$$\Rightarrow B(I) = \begin{bmatrix} 3 & 3 & 1 \\ 4 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix} \quad \text{and} \quad N(I) = \begin{bmatrix} 2 & 4 & 0 \\ 3 & -3 & 1 \\ -1 & 2 & 0 \end{bmatrix}$$

Example: Partition of c and x

Let $n = 6$ and $I = \{5, 3, 2\}$

$$c_B^T = [c_i \mid i \in I] = [c_5, c_3, c_2]$$

$$c_N^T = [c_j \mid j \notin I] = [c_1, c_4, c_6]$$

$$x_B = [x_i \mid i \in I] = [x_5, x_3, x_2]$$

$$x_N = [x_j \mid j \notin I] = [x_1, x_4, x_6]$$

Basic Representation (cont)

Given this partition, we have:

$$\left\{ \begin{array}{l} z = c^T x \\ A x = b \end{array} \right\} \iff \left\{ \begin{array}{l} z = c_B^T x_B + c_N^T x_N \\ B x_B = b - N x_N \end{array} \right.$$

Since B is invertible by construction, this implies that

$$x_B = B^{-1} B x_B = B^{-1}(b - N x_N) = B^{-1}b - B^{-1}N x_N.$$

Substituting this formula into the expression for z we find

$$z = c_B^T x_B + c_N^T x_N = c_B^T B^{-1}b + (c_N^T - c_B^T B^{-1}N)x_N,$$

which may be equivalently rewritten as

$$z = c_B^T B^{-1}b + (c_N - N^T B^{-T} c_B)^T x_N$$

where we use the shorthand notation $B^{-T} = (B^{-1})^T$.

Basic Representation (cont)

Thus, the original system $z = c^T x$, $Ax = b$ is equivalent to the **basic representation**

$$\left. \begin{aligned} z &= c_B^T B^{-1} b + (c_N - N^T B^{-T} c_B)^T x_N \\ x_B &= B^{-1} b - B^{-1} N x_N, \end{aligned} \right\} \quad (*)$$

which expresses z and x_B as **linear functions** of x_N .

Note: By setting $x_N = 0$ in $(*)$ we obtain the **basic solution** $x = (x_B, x_N) = (B^{-1} b, 0)$ with objective value $z = c_B^T B^{-1} b$.

Definition: We call $r = c_N - N^T B^{-T} c_B$ the **reduced cost vector**. This vector characterises the **sensitivity** of the objective function value z with respect to the nonbasic variables x_N .

Example: Basic Representation

Consider the following LP:

$$\min z = 6x_1 + 3x_2 + 4x_3 + 2x_4 - 3x_5 + 4x_6$$

subject to:

$$2x_1 - 1x_2 + 3x_3 + 2x_4 + 3x_5 + 2x_6 + x_7 = 4$$

$$3x_1 + 4x_2 + 2x_3 + 2x_4 + 3x_5 + x_8 = 2$$

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8 \geq 0$$

Example: Basic Representation (cont)

Thus, we are given the following problem data:

$$A = \begin{bmatrix} 2 & -1 & 3 & 2 & 3 & 2 & 1 & 0 \\ 3 & 4 & 2 & 2 & 3 & 0 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad c = \begin{bmatrix} 6 \\ 3 \\ 4 \\ 2 \\ -3 \\ 4 \\ 0 \\ 0 \end{bmatrix}$$

Example: Basic Representation (cont)

Choose $I = \{4, 3\}$. Then, we have

$$B = \begin{bmatrix} 2 & 3 \\ 2 & 2 \end{bmatrix} \Rightarrow B^{-1} = \begin{bmatrix} -1 & \frac{3}{2} \\ 1 & -1 \end{bmatrix},$$

$$N = \begin{bmatrix} 2 & -1 & 3 & 2 & 1 & 0 \\ 3 & 4 & 3 & 0 & 0 & 1 \end{bmatrix},$$

$$c_B^T = \begin{bmatrix} 2 & 4 \end{bmatrix}, \quad c_N^T = \begin{bmatrix} 6 & 3 & -3 & 4 & 0 & 0 \end{bmatrix}.$$

Example: Basic Representation (cont)

Using (*), we find that the **original system**

$$\begin{array}{ccccccccc} z = & 6x_1 & & +3x_2 & +4x_3 & +2x_4 & -3x_5 & +4x_6 & \\ & 2x_1 & -1x_2 & +3x_3 & +2x_4 & +3x_5 & +2x_6 & +x_7 & = 4 \\ & 3x_1 & +4x_2 & +2x_3 & +2x_4 & +3x_5 & & & +x_8 = 2 \end{array}$$

is equivalent to the **basic representation**

$$\begin{array}{ccccccc} z = & 6 & +5x_1 & +9x_2 & & -6x_5 & & -2x_7 & +x_8 \\ x_4 = & -1 & -\frac{5}{2}x_1 & -7x_2 & & -\frac{3}{2}x_5 & +2x_6 & +x_7 & -\frac{3}{2}x_8 \\ x_3 = & 2 & +x_1 & +5x_2 & & & -2x_6 & -x_7 & +x_8 \end{array}$$

The corresponding BS is **not feasible**:

$$(z, x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8) = (6, 0, 0, 2, -1, 0, 0, 0, 0)$$

Importance of Basic Representations

Fix an index set I with $|I| = m$.

- ▶ Assume that
 - ▶ the basis B is invertible and
 - ▶ the corresponding BS with $x_B = B^{-1}b$ and $x_N = 0$ is feasible, i.e., $B^{-1}b \geq 0$.
- ▶ The objective value of this BFS is $z = c_B^T B^{-1}b$.
- ▶ Any other feasible solution satisfies $x_N \geq 0$.
- ▶ The basic representation

$$z = c_B^T B^{-1}b + r^T x_N \quad \text{and} \quad x_B = B^{-1}b - B^{-1}N x_N$$

tells us how z and x_B change when the nonbasic variables increase.

Importance of Basic Representations

In particular, the **reduced cost vector** r enables us to:

- ▶ recognise **whether the current BFS is optimal** (this is the case iff $r \geq 0$; then, no other feasible solution can have a lower objective value than the current BFS);
- ▶ **find a new BFS with a lower objective value** if the current BFS is not optimal (by increasing a nonbasic variable with a negative reduced cost).